📚 e-Business Overview and Models

Brief Overview

This note covering **electronic business** was created from a **101-page PDF**(https://user-uploads-production.api-turbo.ai/b158854d-0554-4ba9-b0c3-5dbbec20ba91/documents/94e8359a-5479-4631-8f32-c88ea705109c.pdf). It gives a snapshot of business fundamentals, e-commerce vs e-business, value chain integration, and e-procurement case studies.

Key Points

- Core **e-business** concepts, tools, and strategic approaches.
- Clear distinction between e-commerce and e-business scopes.
- In-depth look at the value chain and supply-chain technologies.
- Practical examples from e-procurement and ITC e-Choupal initiatives.

📚 COURSE OVERVIEW

- **Objective**: Introduce concepts, tools, and approaches to electronic business for under- and post-graduate students.
- **Key Skills**: Managing businesses in the digital world, enabling functional areas for e-business, leveraging technologies, and applying decision-support tools.

> *"In five years there won't be any Internet companies. All companies will be Internet companies or they will be dead."* – Andy Grove, Intel Chairman (Los Angeles Times, 1999)

📖 What Is Business?

> *Business is the social science of managing people to organize and maintain collective productivity toward accomplishing particular creative and productive goals, usually to generate revenue.* – Wikipedia

> *A business (enterprise, company, firm) is an organizational entity that provides goods or services to consumers. It requires investment and a consistent customer base to*

> *earn profit.* – BusinessDictionary.com

- **Core Elements**
  - Defined **strategy** and **manager** with sales/profit responsibility (Aaker & McLoughlin).
  - Generates **revenue** by delivering **value** to customers and satisfying them.

Purpose of a Business

- **Generate revenue**
- **Deliver value** to customers
- **Satisfy customers**

🎯 Customer Delivered Value (Kotler)

> *Customer-delivered value = Total customer value – Total customer cost.*

- **Total customer value**: Bundle of benefits customers expect from a product/service.
- **Total customer cost**: Bundle of costs customers incur in evaluating, obtaining, using, and disposing of the product/service.

Customer Satisfaction

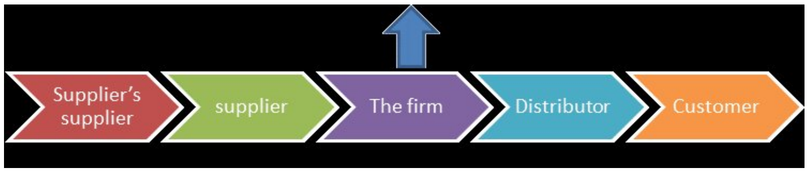
> *Satisfaction = Feeling of pleasure or disappointment from comparing perceived performance with expectation.*

- Satisfied customers → **loyal customers**.
- Loyalty is driven by **high customer value** and a **competitively superior value proposition**.

🔗 Value Chain & Value-Delivery Network (Supply Chain)

> *Primary business processes* – Procurement, Technology Development, HR Development, Firm Infrastructure, Inbound Logistics, Operations, Outbound Logistics, Marketing & Sales, Service.

> *Support activities* – Same as primary processes but enable them.



The diagram visualizes the linear flow of goods/services and the upward feedback arrow that represents information or returns, highlighting inter-organizational dependencies.

🏢 Business vs. Commerce

- **Business**: All organizations that engage in commercial activity.
- **Commerce**: Subset of business focused on the **buying and selling** of goods/services on a large scale.

⏳ Evolution of "e" – From Data Processing to E-Business

Decade	Dominant System	Key Function
1950-60	**Electronic Data Processing (EDP)**	Transaction processing, record keeping, accounting
1960-70	**Management Information Systems (MIS)**	Pre-specified management reports
1970-80	**Decision Support Systems (DSS)**	Interactive, ad-hoc managerial decision support
1980-90	**Strategic & End-User Systems**	Expert systems, strategic information for competitive advantage
1990-present	**E-Commerce & E-Business**	ERP, web-based systems, cloud, mobile, IoT, analytics, big data

---

🌐 ICT for Managing the Value Chain (Historical Perspective)

- Mirrors the evolution above, culminating in **ERP + new enabling technologies** (EDI, RFID, Sensors, IoT, GPS, GIS, Web services, Cloud) for integrated supply-chain operations.

🛒 E-Commerce vs. E-Business

- **E-Commerce**: Buying & selling goods/services via the Internet; includes electronic payment systems.
- **E-Business** (coined by Lou Gerstner, IBM): A **superset** of e-commerce encompassing all online business activities:
  - **Customer servicing**
  - **Partner collaboration**
  - **Internal transactions**

> *In general, e-business means conducting **all** business activities online.*

📊 Defining E-Commerce & E-Business

- **E-Commerce**: Buying, selling, marketing, and servicing products/services over computer networks.
- **E-Business**: Any business process relying on automated information systems, predominantly Web-based.
- **Scope of E-Business**:
  - Electronic purchasing & supply-chain management
  - Order processing
  - Customer service
  - Partner collaboration

📚 Recommended References

- Laudon & Laudon, *Management Information Systems: Managing the Digital Firm*
- Menasce & Almeida, *Scaling for E-Business*
- Meier & Stormer, *eBusiness & eCommerce – Managing the Digital Value Chain*

🔄 Types of E-Business Transactions

Transaction	From	To
C2C	Customer/Consumer	Customer/Consumer
C2B	Customer/Consumer	Business
C2G	Customer/Consumer	Government
B2C	Business	Customer/Consumer
B2B	Business	Business
B2G	Business	Government
G2C	Government	Customer/Consumer
G2B	Government	Business
G2G	Government	Government

Illustrative Cases

- **C2C – eBay**
  - Platform hosted by an intermediary.
  - Issues: payment security, trust.

The image underscores eBay's function as a consumer-to-consumer trading platform.

- **B2C – Dell**
  - Direct sales to consumers via Dell.com; can be intermediary-hosted or owned.
- **C2B – Amazon Reviews**
  - Customers provide feedback that influences business decisions.
- **B2B – Metaljunction.com**
  - Automation of inter-company processes; a marketplace for steel industry participants.
- **G-related examples**:
  - **C2G**: Paying taxes online.
  - **G2C**: Public query portals.
  - **G2B**: License registration.
  - **B2G**: Supplying products to government contracts.
  - **G2G**: Inter-governmental e-payments.

🏭 Integrating Brick-and-Mortar with E-Business

- **Key Enablers**: Supply-chain management, e-procurement, e-marketing, e-selling, CRM, B2B/B2C platforms.
- **Advantages**
  - Better service availability
  - Reduced information-processing cost
  - Faster service delivery
  - Expanded market access
  - Lower initial and operating costs
  - Improved supplier pricing & quality
  - Enhanced product development & scheduling

📈 Business Models on the Web

Model	Core Idea	Typical Example
Brokerage	Connect buyers & sellers, charge fee/commission	eBay, PayPal, Orbitz
Advertising	Provide content & sell ad space	Yahoo!, Google AdWords
Infomediary	Supply market information to participants	DoubleClick, Nielsen
Merchant	Sell goods/services directly	Amazon, Lands' End, Barnes & Noble
Manufacturer (Direct)	Sell directly to end-users, bypass intermediaries	Dell
Affiliate	Drive traffic to merchants, earn performance-based fees	Amazon Associates, DoubleClick
Community	Monetize a user community via ads, donations, or services	Reddit, Open-source projects (Red Hat)
Subscription	Charge periodic fee for access	Netflix, AOL
Utility	Pay-as-you-go based on usage	ISP metered plans, Slashdot

Model Development Steps (Tece, 2010)

1. **Select technologies/features** for product/service.
2. **Determine customer benefits**.
3. **Identify target market segments**.
4. **Confirm revenue streams**.
5. **Design value-capture mechanisms**.
6. **Design value-delivery mechanisms**.
7. **Create value for customers**, convert payments to profit.

🛠️ Innovative E-Business Model: Tata Steel E-Procurement

- **Company Profile**:
  - Contributes >13 % of India’s steel production.
  - FY 2002-03 turnover: **$19.6 bn**, profit: **$2.2 bn**.
- **E-Procurement Timeline**

Year	Milestones
1999-2000	Auto bill payment (IBM), auto-indenting, electronic routing/approval, DSS for stores
2000-2001	SAP ERP rollout, strategic e-procurement analysis, Lotus Notes for supplier email, intranet MRO homepage
2001-2002	SAP-MM stabilization, e-procurement in MRO(P), SAP linkage, reverse auctions, e-routing of import orders, customer complaint portal
2002-2003	Online “Code of Conduct”, supplier satisfaction survey, online bidding for price discovery
Vendor Adoption	Steady increase in number of vendors using the portal (graph data omitted)

- **Key Solutions**
  - **Metaljunction.com** – Industry e-market (SAIL & Tata Steel).

  - **Internal e-Bidding** & **e-Negotiation** platforms.
  - **Online Stock Info System** for VMI suppliers.
- **Benefits**
  - Streamlined purchasing, reduced cycle times, enhanced price transparency, improved supplier engagement.

📂 Instruction for Developing a Case Study

1. **Background & History** – Company overview.
2. **Focus Area** – Procurement, selling, marketing, HR, operations, CRM, supplier or supply-chain management.
3. **Underlying IS** – Describe the information system architecture.
4. **Business Process Re-engineering** – Changes due to e-business adoption.
5. **Perceived Benefits** – Revenue models, cost savings, performance gains.
6. **Success/Failure Stories** – Evidence from implementation.
7. **Analysis & Viewpoint** – Critical assessment.
8. **References** – Minimum 20 credible sources.
9. **Multimedia** – Up to 10 PPT/audio/video assets (optional).

🛠️ E-Procurement Workflow on SAP

> **E-procurement** – an integrated set of processes that automate the acquisition of goods and services, from requisition to payment, using an Enterprise Resource Planning (ERP) system such as SAP.

Core Process Steps

Step	Description	SAP Module / Transaction
1️⃣ **Purchase Requirement (PR)**	Internal department identifies need and creates a requisition.	**MM01** – Create Purchase Requisition
2️⃣ **Request for Quotation (RFQ)**	Procurement issues RFQ to pre-qualified suppliers based on PR details.	**ME41** – Create RFQ
3️⃣ **Supplier Quotations**	Suppliers submit price/terms; quotations are captured electronically.	**ME47** – Enter Quotation

4️⃣ **Opening & Tabulation**	System automatically ranks quotations (price, delivery, quality).	**ME49** – Compare Quotations
5️⃣ **Order Placement**	Winning supplier receives a Purchase Order (PO).	**ME21N** – Create PO
6️⃣ **Goods Receipt (GR)**	Supplier delivers goods; warehouse records receipt.	**MIGO** – Goods Receipt
7️⃣ **Invoice Verification**	Supplier invoice is matched against PO and GR.	**MIRO** – Invoice Receipt
8️⃣ **Payment Processing**	Finance releases payment according to agreed terms.	**F-53** – Post Outgoing Payment
9️⃣ **Reverse Auction (optional)**	Time-bound electronic bidding to drive price down.	Integrated with **ME41/ME49** workflow
🔟 **Business Process Outsourcing (BPO)**	Certain steps (e.g., RFQ handling) can be outsourced to partner firms.	Managed via **SAP BPO** interfaces

Key advantage: End-to-end visibility, reduced manual effort, and faster cycle times (see performance metrics below).

📊 Performance Results from Tata Steel's e-Procurement

> **Strategic sourcing savings** – cumulative cost reductions achieved by moving from manual to electronic processes.

Metric	1999-00	2000-01	2001-02	2002-03	2003-04
Average Order Lead Time (days)	97-98	98-99	99-00	00-01	01-02
Strategic Sourcing	0	50	100	150	200

Savings (₹ crore)					
Inventory Value (₹ crore)	0	50	100	150	200 (target 250)
Cumulative Inventory Reduction	–	–	–	**30 %**	**45 %**

Interpretation: Lead-time compression and inventory draw-down directly contributed to the observed savings.

🌾 ITC e-Choupal Initiative – End-to-End View

> **e-Choupal** – a farmer-centric ICT platform that re-engineers soybean procurement by delivering real-time market and agronomic information to rural producers.

1️⃣ Origin & Management Philosophy

- Launched by ITC Limited's International Business Division (IBD) under CEO **S. Sivakumar**.
- **Principle 1:** *Re-engineer, don't reconstruct* – focus on process redesign rather than merely adding technology.
- **Principle 2:** *Address the whole, not just a part* – integrate information, logistics, and finance across the supply chain.

2️⃣ Technical Architecture

Component	Specification
Hardware Kit	PC (Windows/Intel), multimedia kit, dot-matrix printer, UPS + solar backup
Connectivity	Telephone line (28.8–36 kbps) or VSAT (satellite)
Cost	₹ 170,000 (~$3,762) for hardware; additional ₹ 100,000 (~$2,213) for training & deployment

Data Flow	Field data uploaded via VSAT → Bangalore (ITC Infotech) → Bhopal (portal maintenance) → Farmers (web site)
Portal URL	www.soyachoupal.com (dedicated to soybean growers)

3️⃣ Portal Feature Set

- **Weather** – localized forecasts.
- **Best Practices** – agronomic guidelines.
- **Crop Information** – varietal details, pest alerts.
- **Market Information** – domestic & international price listings, trend commentary.
- **Agri-Inputs** – directory of certified input manufacturers.
- **Alerts** – region-specific advisories (e.g., disease outbreaks).
- **Soil & Water Testing** – sample submission, online results.
- **News** – industry updates.

> *Feature development involved continuous farmer participation, ensuring relevance and usability.*

4️⃣ Kiosk Establishment Process

1. **Mapping** – identify internet-enabled telephone exchanges.
2. **Village Selection** – choose agrarian-active villages.
3. **"Prathinidhi" Identification** – appoint a progressive farmer as kiosk manager.

Infrastructure: PC, UPS, printer, telephone, internet line, earthing.

Training: "Prathinidhi" + 10-15 villagers receive hands-on PC and portal instruction from trained operators.

5️⃣ Redesign of the Soybean Supply Chain

Element	Traditional Mandi System	e-Supply Chain (e-Choupal)
Price Discovery	Dependence on agents; possible manipulation	Real-time market prices via portal
Information Flow	One-way, delayed	Two-way, instantaneous (feedback, alerts)

Weighing & Payment	Manual scales, prone to tampering	Electronic scales; transparent payment
Logistics	Bagged handling → multiple unloads	Direct hub collection; unbagged, faster unload
Transaction Costs	High (agent commissions, loans)	Lower (digital transaction, reduced intermediaries)
Quality Control	Arbitrary inspections	Standardized grading at hub, farmer-partner status

6️⃣ Benefits for Stakeholders

Farmers

- **2.5 % higher price** (≈ $6 / tonne) due to better price information.
- **Transaction time** reduced to 2-3 hours versus days at mandis.
- Accurate electronic weighing eliminates manual bias.
- No bagging losses; lower handling waste.
- Access to cheaper inputs (seed, fertilizer) via the portal.
- Ability to improve yields using agronomic advice.

Processors

- **Elimination of intermediaries** → lower transaction costs.
- Direct engagement builds trust; farmers viewed as *business partners*.
- Higher product quality and reduced post-harvest losses.
- Influence over farming practices (e.g., adoption of better inputs).

Overall Cost-Benefit

- **Net gain**: higher farmer realization + lower processor costs → improved profit margins for both parties.
- **Systemic effect**: increased soybean area under cultivation, yet reduced reliance on traditional mandis.

📈 Comparative Analysis: Traditional Mandi vs. e-Supply Chain

Dimension	Traditional Mandi	e-Supply Chain (e-Choupal)

Information Accuracy	Low; agents control price data	High; real-time, verified market data
Transaction Speed	Days to weeks	Hours
Weight Verification	Manual, manipulable	Electronic, tamper-proof
Handling Losses	Multiple bagging/unbagging steps	Direct hub unload, minimal handling
Farmer’s Role	Passive seller	Active partner with decision-making tools
Cost Structure	High (agent fees, loan interest)	Low (digital platform, reduced middlemen)
Quality Assurance	Arbitrary, buyer-biased	Standardized grading at hub

---

🔄 Business Process Outsourcing (BPO) & Supplier Partnerships in E-Procurement

- **Outsourced Activities**: RFQ preparation, quotation analysis, reverse-auction management, and e-negotiation can be handled by specialized BPO firms, allowing the buying organization to focus on strategic sourcing.
- **Partnership Models**
    - **Strategic Supplier Portals** (e.g., **Metaljunction.com**) for market-wide scanning when cost or quality is critical.
    - **Volume-Based Agreements** – high-volume buyers negotiate bulk discounts via electronic contracts.
    - **Portfolio-Driven e-Procurement** – categorizing spend into low, medium, high risk/volume to apply appropriate digital tools.

> *Effective BPO integration reduces order lead time and enhances sourcing agility, as reflected in the Tata Steel performance chart.*

📚 References

- Teece, D.J. (2010). *Business models, business strategy and innovation.* Long Range Planning, 43(2), 172-194.

- ITC Limited, *e-Choupal case study*, Department of IE&M (lecture material).
- Tata Steel, *e-Procurement implementation data* (lecture material).
- Additional sources listed in earlier sections (dot-com history, etc.).